

JavaScript Topics (Core + Advanced)

These are the foundation before you touch React seriously:

1 Basics & Fundamentals

- Variables (`var`, `let`, `const`)
- Data Types (Primitive & Reference)
- Operators (Arithmetic, Logical, Comparison)
- Conditional Statements (`if`, `else`, `switch`)
- Loops (`for`, `while`, `for...of`, `forEach`)
- Functions (Declaration, Expression, Arrow Functions)
- Scope (Global, Local, Block, Lexical)
- Hoisting
- Template Literals
- Type Conversion (Implicit & Explicit)

2 Arrays & Objects

- Array Methods (`map`, `filter`, `reduce`, `find`, `some`, `every`, `sort`)
- Object Methods (`keys`, `values`, `entries`)
- Destructuring Arrays & Objects
- Spread & Rest Operators
- JSON (`parse`, `stringify`)

3 DOM & Events

- Selecting Elements (`querySelector`, `getElementById`)
- Event Handling (`onclick`, `addEventListener`)
- Event Bubbling & Delegation
- Creating & Removing Elements (DOM Manipulation)

4 Advanced JavaScript

- ES6+ Features (`let`, `const`, arrow functions, classes, modules, promises)
- Callbacks & Promises
- Async/Await
- Fetch API (GET, POST requests)
- Closures
- Higher-Order Functions
- `this` keyword & Binding
- Prototypes & Inheritance
- Error Handling (`try`, `catch`, `finally`)
- LocalStorage & SessionStorage

5 Miscellaneous

- Debouncing & Throttling
- Event Loop & Call Stack
- JavaScript Modules (`import`, `export`)
- Regular Expressions (Regex)
- JavaScript best practices & clean code

React Topics (Core + Advanced)

1 Basics

- What is React? Why use it?
- JSX
- Components (Functional vs Class-based)
- Props & Prop Drilling
- State Management (`useState`)
- Conditional Rendering
- Lists & Keys (`map` with JSX)
- Event Handling in React

2 Hooks

- `useState`
- `useEffect` (side effects, dependencies)
- `useRef`
- `useContext`
- `useReducer` (mini Redux)
- Custom Hooks

3 React Essentials

- Forms & Controlled Components

- Handling Events
- React Router (Navigation, `useNavigate`)
- React Fragments
- React Strict Mode
- React Refs (Forwarding refs)

4 Advanced React

- Context API (Global state without Redux)
- React Performance Optimization (`memo`, `useMemo`, `useCallback`)
- Error Boundaries
- Lazy Loading & Code Splitting
- Portals
- Higher-Order Components (HOCs)
- Render Props

5 State Management Libraries

- Redux (Actions, Reducers, Store, Middleware)
- Redux Toolkit
- Zustand, Recoil (alternatives)

6 APIs & Data Fetching

- Fetch API with React
- Axios in React

- Custom hooks for API calls

7 Styling in React

- Inline Styling
- CSS Modules
- Styled Components
- Tailwind CSS
- Framer Motion (for animations)

8 Testing

- Jest
- React Testing Library

9 Deployment

- Build & Deploy with Vite or Create React App
- Hosting on Netlify, Vercel, GitHub Pages